

Cell Ratio Derivation for 6T SRAM Read Stability

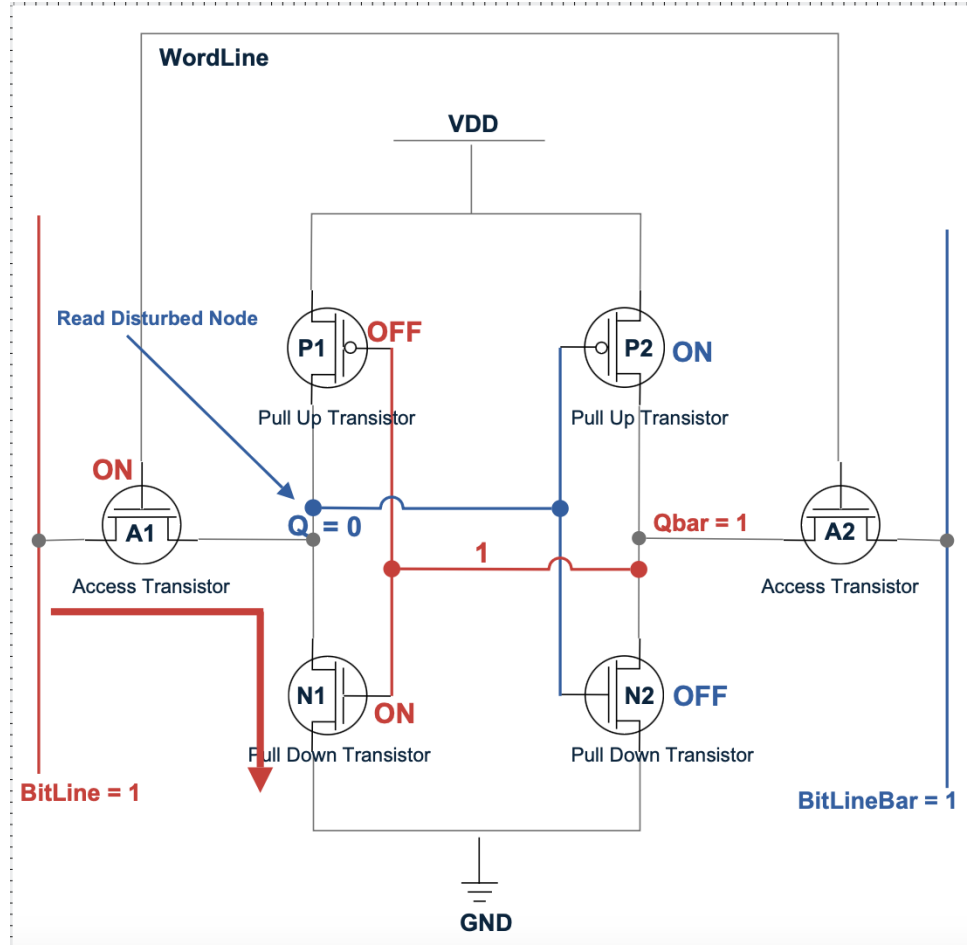


Figure 1: Read stability analysis of the 6T SRAM cell during a read operation.

Read Stability Analysis

Assume the SRAM cell initially stores

$$Q = 0, \quad \bar{Q} = 1$$

During the read operation,

$$WL = V_{DD}, \quad BL = BLB = V_{DD}$$

Let the internal storage node Q rise to an unknown voltage V_1 .

Access NMOS (A1)

$$V_G = V_D = V_{DD}, \quad V_S = V_1$$

$$V_{GS} = V_{DS} = V_{DD} - V_1$$

Since

$$V_{DS} \geq V_{GS} - V_{TN},$$

the Access transistor operates in saturation.

$$I_{A1} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_{A1} (V_{DD} - V_1 - V_{TN})^2$$

Pull-Down NMOS (N1)

$$V_G = V_{DD}, \quad V_D = V_1, \quad V_S = 0$$

$$V_{GS} = V_{DD}, \quad V_{DS} = V_1$$

Since

$$V_{DS} < V_{GS} - V_{TN},$$

the Pull-Down transistor operates in the linear region.

$$I_{N1} = \mu_n C_{ox} \left(\frac{W}{L} \right)_{N1} \left[(V_{DD} - V_{TN})V_1 - \frac{V_1^2}{2} \right]$$

Applying Kirchhoff's Current Law (KCL)

At node Q ,

$$I_{A1} = I_{N1}$$

Substituting the drain current equations,

$$\frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_{A1} (V_{DD} - V_1 - V_{TN})^2 = \mu_n C_{ox} \left(\frac{W}{L} \right)_{N1} \left[(V_{DD} - V_{TN})V_1 - \frac{V_1^2}{2} \right]$$

Cancelling the common term $\mu_n C_{ox}$,

$$\frac{1}{2} \left(\frac{W}{L} \right)_{A1} (V_{DD} - V_1 - V_{TN})^2 = \left(\frac{W}{L} \right)_{N1} \left[(V_{DD} - V_{TN})V_1 - \frac{V_1^2}{2} \right]$$

Multiplying both sides by 2,

$$\left(\frac{W}{L} \right)_{A1} (V_{DD} - V_1 - V_{TN})^2 = \left(\frac{W}{L} \right)_{N1} [2(V_{DD} - V_{TN})V_1 - V_1^2]$$

Hence,

$$\boxed{\frac{\left(\frac{W}{L} \right)_{A1}}{\left(\frac{W}{L} \right)_{N1}} = \frac{2(V_{DD} - V_{TN})V_1 - V_1^2}{(V_{DD} - V_1 - V_{TN})^2}}$$

Physical Interpretation

For reliable read operation, the Pull-Down NMOS transistor must be stronger than the Access NMOS transistor. During a read operation, the Word Line (WL) is asserted HIGH, turning ON the access transistor while both bitlines are precharged to V_{DD} . Since the storage node Q initially stores logic '0', the precharged bitline attempts to raise the voltage at node Q through the access transistor. If this voltage exceeds the inverter switching threshold, the cross-coupled inverters may flip their stored state, resulting in a read disturb and incorrect data being read.

Therefore, the Pull-Down transistor is intentionally sized stronger than the Access transistor so that it can effectively hold node Q at logic '0' during the read operation. Hence,

$$\boxed{\frac{\left(\frac{W}{L} \right)_{A1}}{\left(\frac{W}{L} \right)_{N1}} < \frac{2(V_{DD} - V_{TN})V_1 - V_1^2}{(V_{DD} - V_1 - V_{TN})^2}}$$

Equivalently,

$$\boxed{CR = \frac{(W/L)_{N1}}{(W/L)_{A1}} > 1}$$

In practical SRAM designs,

$$\boxed{CR \approx 1.5 - 2}$$

to ensure adequate read stability.